



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For



CompTIA Certified Linux+ Training (XK0-006)

TGS Ref No: TGS-2024048316

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	2 Jul 2026	Initial release — CompTIA Linux+ XK0-006 V8 2-day lesson plan aligned to the 30 labs, the five exam domains and the slide deck (slide numbers cited).	Dr. Alfred Ang

TABLE OF CONTENTS

Right-click and choose “Update Field”, or press F9, to build the table of contents.

Course Information

Course Title	CompTIA Certified Linux+ Training (XK0-006)
WSQ Course Reference	TGS-2024048316
Certification Exam	CompTIA Linux+ XK0-006 V8
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 8 training hours per day (16 hours)
Daily Timing	9:00 am – 6:00 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on labs on the free KillerCoda Ubuntu Playground
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- Explain core Linux concepts — the boot process, the Filesystem Hierarchy Standard, device management and virtualization — and manage storage, networking and shell operations.
- Manage files, local user and group accounts, processes and jobs, software packages, systemd services and containerized applications on a Linux server.
- Apply Linux security: authentication/authorization (sudo, PAM), firewalls, OS and account hardening, cryptography and compliance/audit procedures.
- Automate administration with Ansible, Bash and Python, apply Git version control, and use AI assistants responsibly and securely.
- Monitor a Linux system and analyze and troubleshoot hardware, storage, network, security and performance issues using the right diagnostic tools.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book — aligned to the course slides.
- Practical Performance (PP) — hands-on Linux tasks, 1 hour, open book — aligned to the course labs.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The final assessment is conducted on Day 2 from 4:00 pm (WA 4:00–5:00, PP 5:00–6:00).
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- Courseware and the assessment are on the LMS — <https://lms-tms.tertiaryinfotech.com/>.

Exam Domain Coverage

The course maps 1:1 to the CompTIA Linux+ XK0-006 exam blueprint. Each domain, its exam weighting, the labs that cover it, and the deck slides:

Domain	Exam Weight	Labs	Deck Slides
1. System Management	23%	1–7	21–89
2. Services and User Management	20%	8–13	87–148
3. Security	18%	14–19	146–207
4. Automation, Orchestration, and Scripting	17%	20–24	205–256
5. Troubleshooting	22%	25–30	254–319

Daily Schedule

Each training day runs 9:00 am – 6:00 pm and delivers 8 instructional hours (1-hour lunch; tea breaks counted within training time). The Slides column cites the deck page range the trainer runs for each teaching session.

Day 1 — System Management & Services / User Management

Time	Duration	Topic / Activity	Method	Slides
9:00–9:15	15 min	Welcome, mandatory digital attendance (AM), trainer & learner introduction	Admin	—
9:15–9:30	15 min	Course overview, learning outcomes and ground rules	Admin	—
9:30–11:45	135 min	Domain 1 — System Management (23%): boot & FHS, kernel modules, LVM storage, networking, shell operations. Hands-on Labs 1–5	Lecture + Hands-on Lab	21–66
11:45–12:45	60 min	Lunch break	Break	—
12:45–14:15	90 min	Domain 1 (cont.): backup & restore, virtualization. Hands-on Labs 6–7	Lecture + Hands-on Lab	67–89
14:15–14:25	10 min	Digital attendance (PM)	Admin	—
14:25–17:15	170 min	Domain 2 — Services and User Management (20%): files, accounts, processes, packages,	Lecture + Hands-on Lab	87–148

		systemd, containers. Hands-on Labs 8–13		
17:15–17:45	30 min	Domain 1–2 recap, Q&A and consolidation	Discussion	—
17:45–18:00	15 min	Day 1 review, preview of Day 2 and PM digital attendance	Discussion	—

Total training time: 480 minutes (8 hours).

Day 2 — Security, Automation, Troubleshooting & Final Assessment

Time	Duration	Topic / Activity	Method	Slides
9:00–9:10	10 min	Mandatory digital attendance (AM) and Day 1 recap	Admin	—
9:10–11:45	155 min	Domain 3 — Security (18%): AAA/sudo/PAM, firewalls, OS & account hardening, cryptography, compliance & audit. Hands-on Labs 14–19	Lecture + Hands-on Lab	146–207
11:45–12:45	60 min	Lunch break	Break	—
12:45–14:35	110 min	Domain 4 — Automation, Orchestration & Scripting (17%): Ansible, Bash, Python, Git, responsible AI. Hands-on Labs 20–24	Lecture + Hands-on Lab	205–256
14:35–14:45	10 min	Digital attendance (PM)	Admin	—
14:45–15:35	50 min	Domain 5 — Troubleshooting (22%): monitoring, storage, network, security, performance. Hands-on Labs 25–29	Lecture + Hands-on Lab	254–301
15:35–15:55	20 min	Capstone — end-to-end server build & triage. Hands-on Lab 30	Lecture + Hands-on Lab	302–319
15:55–16:00	5 min	Revision · Briefing for Assessment · Course feedback &	Assessment	—

		TRAQOM survey		
16:00–17:00	60 min	Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book	Assessment	—
17:00–18:00	60 min	Practical Performance (PP) — hands-on Linux tasks, 1 hour, open book. PM digital attendance; submit answers on the LMS	Assessment	—

Total training time: 480 minutes (8 hours).

Topic-by-Topic Breakdown

Each lab below cites the deck slide of its overview so the trainer can jump straight to it.

Domain 1 — System Management (23%) · Slides 21–89

Lab 1 — Boot Process & Filesystem Hierarchy [Obj 1.1] · Slide 24

Inspect a running Linux system's boot configuration, kernel command line, initramfs, and the standard Filesystem Hierarchy to understand how GRUB hands off to the kernel and where FHS directories live.

Lab 2 — Kernel Modules & Device Management [Obj 1.2] · Slide 32

List, inspect, load, and unload kernel modules and use the standard hardware-enumeration tools to understand modprobe vs insmod, module dependencies, and reading dmesg.

Lab 3 — Storage with LVM, Partitions & Filesystems [Obj 1.3] · Slide 41

Build a complete LVM stack on loopback-backed disks, format with ext4/xfs/btrfs, mount with hardening options, persist in fstab, and grow a filesystem online.

Lab 4 — Network Configuration [Obj 1.4] · Slide 49

Inspect interfaces, addresses, and routes with ip; walk the name-resolution stack; validate a netplan YAML; and use the standard diagnostic toolbox to read a Linux network stack top-to-bottom.

Lab 5 — Shell Operations & Text Processing [Obj 1.5] · Slide 58

Work with shell environment variables, login vs interactive startup files, every common redirection form, and the classic text-processing pipeline to assemble non-trivial one-liners.

Lab 6 — Backup & Restore [Obj 1.6] · Slide 67

Create tar/cpio/dd archives with gzip/bzip2/xz compression, image and recover disks with dd and ddrescue, mirror trees with rsync, and verify integrity with sha256sum.

Lab 7 — Virtualization with QEMU/KVM & libvirt [Obj 1.7] · Slide 76

Install QEMU and libvirt, manipulate virtual disk images with qemu-img, list libvirt domains/networks with virsh, and boot a tiny guest under QEMU to discuss hypervisors, image formats, snapshots, and network modes.

Domain 2 — Services and User Management (20%) · Slides 87–148

Lab 8 — File & Directory Management [Obj 2.1] · Slide 90

Practice core file and directory operations: navigating, creating, copying, searching, linking, and inspecting special device files with standard Linux utilities.

Lab 9 — Local Account & Group Management [Obj 2.2] · Slide 99

Create and manage local users and groups, set passwords and aging policies, change shells, and distinguish user, system, and service accounts.

Lab 10 — Processes, Jobs & Scheduling [Obj 2.3] · Slide 108

Inspect processes, manipulate shell jobs, adjust priorities, send signals, and schedule tasks with at, cron, and anacron.

Lab 11 — Software & Package Management [Obj 2.4] · Slide 117

Manage Debian packages with apt/dpkg, add a signed third-party repo, run dnf/rpm in a container, and use language and sandboxed package managers.

Lab 12 — Service Management with systemd [Obj 2.5] · Slide 126

Inspect and control systemd units, author custom service/timer/mount units, analyze boot performance, and configure host/time/DNS via systemd helpers.

Lab 13 — Containers with Docker & Podman [Obj 2.6] · Slide 135

Install Docker and Podman, pull/run images, manage volumes and networks, build an image from a Dockerfile, and contrast rootful vs rootless engines.

Domain 3 — Security (18%) · Slides 146–207

Lab 14 — AAA: sudo, PAM, and Polkit [Obj 3.1] · Slide 149

Configure sudo authorization, inspect PAM authentication stacks, and enable auditd accounting to record security events. Finish with a conceptual look at centralized identity via LDAP, Kerberos, and SSSD.

Lab 15 — Firewalls: ufw, firewalld, iptables, nftables, ipset [Obj 3.2] · Slide 158

Configure firewalls at every layer of the Linux netfilter stack, from the ufw wrapper down to iptables and nftables, and understand stateful inspection. Explore zones, rich rules, ipset, and SNAT/DNAT.

Lab 16 — OS Hardening: Permissions, SELinux, SSH, Fail2ban [Obj 3.3] · Slide 167

Apply layered OS hardening from file permissions and immutability through SELinux labeling, then harden OpenSSH and add brute-force protection. Inventory SUID binaries and legacy cleartext services.

Lab 17 — Account Hardening [Obj 3.4] · Slide 176

Harden user accounts through password quality and aging policies, brute-force lockouts, and restricted or non-interactive shells. Explore TOTP MFA and Have I Been Pwned k-anonymity lookups.

Lab 18 — Cryptography: GPG, LUKS, OpenSSL, WireGuard [Obj 3.5] · Slide 185

Practice core Linux cryptography: GPG key generation and signing, LUKS2 disk encryption, OpenSSL certificates, hashing and HMAC, and WireGuard keys. Learn to identify weak TLS versions and ciphers.

Lab 19 — Compliance, Auditing, and File Integrity [Obj 3.6] · Slide 194

Establish file-integrity monitoring and run compliance and audit tooling to detect tampering, rootkits, and misconfigurations. Cover secure deletion, login banners, and OpenSCAP/CIS benchmarks.

Domain 4 — Automation, Orchestration, and Scripting (17%) · Slides 205–256

Lab 20 — Infrastructure as Code with Ansible [Obj 4.1] · Slide 208

Learn Infrastructure as Code using Ansible as the primary agentless configuration management tool, and recognize the wider IaC ecosystem (Puppet, OpenTofu, cloud-init, Kubernetes ConfigMaps). Build an idempotent playbook that deploys and configures nginx.

Lab 21 — Bash Scripting [Obj 4.2] · Slide 216

Practice core Bash scripting constructs (variables, conditionals, loops, functions, option parsing, regex, file tests) and assemble a defensive backup script, then audit it with shellcheck.

Lab 22 — Python for System Administration [Obj 4.3] · Slide 225

Use Python for common Linux admin tasks: create a virtual environment, exercise core data types, walk the filesystem, parse JSON, call a REST API with requests, and enforce PEP 8 with black and flake8.

Lab 23 — Git Version Control [Obj 4.4] · Slide 234

Work through Git end to end: init, commit, branch, merge, resolve conflicts, stash, tag, rewrite history with reset and rebase, and push/pull against a local bare remote.

Lab 24 — Responsible AI Use for Linux Administrators [Obj 4.5] · Slide 243

Adopt a verify-before-paste workflow for AI-generated Linux commands: redact secrets before sending prompts to hosted models, lint and sandbox suggestions, weigh local vs hosted models, and draft a corporate AI usage policy.

Domain 5 — Troubleshooting (22%) · Slides 254–319

Lab 25 — Monitoring Concepts and Tools [Obj 5.1] · Slide 257

Learn the monitoring vocabulary (SLA/SLO/SLI) and stand up a working monitoring stack to produce metrics, alerts, webhooks, health checks, and aggregated logs.

Lab 26 — Troubleshooting Hardware, Storage, and OS Issues [Obj 5.2] · Slide 266

Deliberately inject common OS and storage failures then diagnose and repair each, keeping every breakage contained to loopback devices, throwaway units, or subshells.

Lab 27 — Troubleshooting Network Connectivity [Obj 5.3] · Slide 275

Break and repair the common network faults (DNS, routing, firewall, MTU, IP, link, dual-stack) while practicing the standard L3/L4 triage toolkit.

Lab 28 — Troubleshooting Security Issues [Obj 5.4] · Slide 284

Diagnose and repair security faults spanning SELinux contexts, POSIX ACLs, certificate trust, account lockout, broken repos, and weak TLS, using a Rocky container for the SELinux portion.

Lab 29 — Troubleshooting Performance Issues [Obj 5.5] · Slide 293

Stress CPU, memory, disk I/O, and network in turn, then walk the universal performance loop: baseline, spike, identify the bottleneck with the right counter, and resolve it.

Lab 30 — Capstone: End-to-End Server Build and Triage [Obj Capstone] · Slide 302

Integrate Objectives 1-5 into one coherent build: a hardened admin account, LVM storage, nginx with a systemd override, a locked firewall, a health-check timer, and git-versioned config, then inject a fault and triage it end-to-end.

Resources Required

- A laptop with a modern web browser and reliable internet access.
- Free Killercode Ubuntu Playground — <https://killercode.com/playgrounds/scenario/ubuntu> (no local install required).
- Course slide deck and Learner Guide (downloaded from the LMS).
- Tertiary Infotech LMS account — <https://lms-tms.tertiaryinfotech.com/> (courseware download and assessment).
- CompTIA Linux+ XK0-006 V8 Exam Objectives (reference blueprint).